JAN-O-592

12 MAY 1948

JOINT ARMY-NAVY SPECIFICATION

OINTMENT, PROTECTIVE, M5

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

U. S. Army
97-54-351
5 July 1945

A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATION, AND DRAWINGS

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-P-120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.
JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN-P-188—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.
JAN-P-140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.
JAN-M-560—Magnesium Stearate (Ointment Type).
JAN-T-561—Triacetin, High-Purity (Glyceryl Triacetate).
JAN-P-564—Pigment, Synthetic, Red (Ointment Type).
JAN-P-567—Pigment, Synthetic, Black (Ointment Type).
JAN-C-590—Cellulose-Acetate-Butyrate (For Use in Protective Ointment).
JAN-O-591—Ointment-Kits, Protective, M5.
JAN-T-602—Titanium-Dioxide (Ointment Type).
JAN-S-603—S-330.

751467—48—(C-283)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

FEDERAL SPECIFICATIONS

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
LLL–B–631—Boxes; Fiber, Corrugated (For Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (For Domestic Shipment).

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[1]

A–3. *Drawings.*—The following drawing, and subsidiary drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

CHEMICAL CORPS DRAWING

E5–43–10—Ointment, Protective, M5—Assembly and Bill of Material.

B. GRADE

B–1. This specification covers one grade of protective ointment, as hereinafter specified.

C. MATERIAL AND WORKMANSHIP

C–1. *Materials.*—The materials used in compounding the protective ointment shall be as specified in paragraph E–1. No materials (including water) other than those specified in paragraph E–1 shall be used.

C–2. *Workmanship.*—The ointment shall be compounded in a manner consistent with the best pharmaceutical practice.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Composition.*—The protective ointment, at the time of tubing, shall be a smooth, homogeneous mixture of the ingredients in the proportions indicated in Table I.[2]

---

[1] Applicable only to Navy purchases.

[2] The solution of cellulose acetate butyrate in triacetin is impeded by the presence of moisture in either material. Ointment consistency is affected by the total moisture content of the finished ointment. Stability is adversely affected with increasing moisture content, therefore the addition of moisture to the finished ointment or to any of its components must be avoided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

TABLE I.—*Composition.*

Ingredients	Specification	Parts by weight
Cellulose acetate butyrate*	JAN-C-590	4.0 ± 0.1
Magnesium stearate	JAN-M-560	9.0 ± 0.2
Mixed pigments	(See par. E-2)	2.0 ± 0.02
S-330	JAN-S-603	25.0 ± 0.5
Titanium dioxide	JAN-T-602	9.0 ± 0.2
Triacetin	JAN-T-561	51.0 ± 2.0

* The cellulose acetate butyrate shall contain not more than 2.0 percent of moisture at the time of compounding the ointment. Cellulose acetate butyrate containing more than 2.0 percent moisture shall be dried, immediately prior to use, at 110° to 120°F. to reduce its moisture content to 2.0 percent or less.

E–2. *Mixed pigments.*—The mixed pigments shall consist of a uniform mixture of 7.00 ± 0.10 parts of black pigment, complying with the requirements of Joint Army-Navy Specification JAN–P–567, and 1.00 ± 0.02 parts of red pigment, complying with the requirements of Joint Army-Navy Specification JAN–P–564.

E–3. *Manufacture.*—The ointment shall be compounded only from ingredients which comply with the requirements of the applicable specifications. Overages from one lot of S–330 may be blended with part of the next lot of S–330 used, provided the blended S–330 is used to manufacture one of the next two lots of ointment.

E–4. *Active chlorine content.*—The active chlorine content of the ointment shall be 7.75 ± 0.50 percent.

E–5. *Consistency.*—The finished ointment shall have a consistency of 55.0 ± 7.0 when tested as prescribed in paragraph F–6.[3]

E–6. *Heat stability.*—

E–6a. *Loss of active chlorine.*—The decrease in active chlorine content of the finished ointment shall be not more than 10 percent of the original, as determined by the method prescribed in paragraph F–5, after being treated as prescribed in paragraph F–7a.

E–6b. *Change in consistency.*—The consistency of the finished ointment, after being treated as prescribed in paragraph F–7a, shall be 58.5 ± 8.5.

E–7. *Tubing.*—

E–7a. *Cleaning.*—The interior of each wax-lined, lead-alloy tube (Cml. C. Dwg. B68–7–5) shall be substantially free from dust at the time of filling.[4]

[3] Consistency of the finished ointment may be varied by varying the triacetin content of the ointment within the tolerances prescribed in paragraph E–1.

[4] The empty tubes may be conveniently freed of dust by means of suction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

E–7b. *Filling.*—Each tube shall be filled with 19 ± 1 gm., net weight, of the finished ointment and the tube closed as shown on Chemical Corps Drawing E5–43–10.

E–7c. *Marking.*—The crimped portion of each filled tube shall be marked as prescribed on Chemical Corps Drawing E5–43–10.

E–8. *Leakage.*—There shall be no leakage of ointment from the filled and closed tubes.

E–9. *Grit.*—The finished ointment shall contain no grit.

E–10. *Texture.*—The texture of the finished ointment shall be approximately equivalent to that obtained by passing the ointment once through a roller mill having a tight setting.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. The material and work shall be subject at all times to inspection at the place of manufacture, and every facility shall be afforded authorized Government inspectors for the prosecution of their work.

F–2. *Size of lot.*—Unless otherwise specified in the contract or order, a lot shall consist of a single manufacturing batch of ointment. (See par. H–3.)

F–3. *Sampling.*—Ten consecutive finished tubes shall be selected from each lot of ointment for test purposes. The tubes shall be taken near the middle of the filling operation. The 10 tubes shall be divided as follows (see par. H–4):

Four tubes for active chlorine test. (See par. F–5.)
Five tubes for heat-stability tests. (See par. F–7.)
One tube for consistency test. (See par. F–6.)

F–4. *Manufacture.*—

F–4a. *Composition.*—The inspector shall assure himself, by personal observation or from suitable records or affidavits, that the ointment is manufactured from the ingredients in the prescribed proportions, and in accordance with the restrictions set forth in paragraphs C–1, E–1, and E–3.

F–4b. *Pigments.*—The inspector shall assure himself that the color pigments were mixed in the proper proportions.

F–4c. *Texture.*—The inspector shall ascertain that the finished ointment complies with the texture requirements. (A sample ointment, which has been passed once through a roller mill with a tight setting may be used for comparison.) If, in the process of manufacture, the ointment was passed through a roller mill with a tight setting, the ointment shall be considered as meeting the requirement specified in paragraph E–10.

F–5. *Active chlorine content of original sample.*—

F–5a. *Preparation of samples.*—Open the four tubes, selected for the active chlorine test, along the entire length of the tube. Transfer the contents of all four tubes to a beaker and mix thoroughly. When the mass is homogeneous, weigh out, on a small square of "weighing paper" (glassine or equal), approximately 1.000 gm. of the ointment.

[JAN-O-591]

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

F–5b. *Test.*—Place 25 ml. of chloroform[5] in a 250-ml., glass-stoppered Erlenmeyer flask. Introduce the paper square and sample into the flask, stopper, and disperse the sample by swirling the solvent. Immediately before titrating add, all at once, 20 ml. of 5-percent potassium iodide solution followed immediately by 10 ml. of glacial acetic acid, quickly stopper the flask and shake thoroughly.[7] Cool the flask to below room temperature in a cold water bath before opening, then remove the stopper, and wash down the stopper and sides of the flask with distilled water. Titrate immediately with 0.1*N* sodium thiosulfate solution, swirling the flask constantly to prevent local excess of thiosulfate. When the aqueous layer becomes pale yellow, stopper the flask and shake vigorously before continuing the titration. (Do not permit an excess of sodium thiosulfate to be present in the aqueous layer while there is still any iodine dissolved in the chloroform.) When most of the iodine has been titrated, add 2 ml. of 1-percent starch solution and continue the titration to the end point, shaking after each addition near the end point. Run a blank determination in parallel with the test using the same weight of "weighing paper" and the same quantities of the reagents used in the test, following the above procedure. If no blank titre is obtained, test the reagents, especially the chloroform, for the presence of reducing matter. Reject any reagent which contains reducing matter and repeat the determination using an acceptable reagent. Subtract the blank titre from that required for the titration of the active chlorine and use the difference in the following calculation:

$$\text{Percentage active chlorine} = \frac{1.773\ AN}{W}$$

where:

A = ml. sodium thiosulfate used in titration
N = normality of sodium thiosulfate solution
W = weight of sample.

F–5c. *Rejection.*—If the composite sample of four tubes fails to comply with the requirements of paragraph E–4, the entire lot shall be rejected.

F–6. *Consistency.*—Between manufacture and test the sample shall be stored at room temperature (81°C., max.).

F–6a. *Apparatus.*—Two pieces of smooth-edged, plate glass, approximately 9 inches by 9 inches by ¼ inch, are required. (It is convenient to place the plates in a shallow tray.) Weigh the top glass plate and prepare a weight such that the total weight of the plate and weight is 2000 gm. (It is convenient to prepare this weight with a circular base 5 cm. or less in diameter.) Mark the under side of the bottom glass plate, conveniently with India ink,

[5] The chloroform shall be washed three times by shaking in a separatory funnel, with half its volume of water each time, and separating. After the third washing, pass the separated chloroform through a dry, fluted, filter paper.

[7] Acidified potassium iodide solution is easily oxidized, by the oxygen in the air, with the formation of free iodine.

[JAN-O-592]

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

as follows: Draw two diagonal lines across the plate so that they intersect at right angles in the center of the plate. Draw a circle 5 cm. in diameter, using the intersection of the lines as a center. Draw two more straight lines across the plate to divide the circle into eight equal sectors of 45° each.

F–6b. *Determination.*—The sample and plates shall be maintained at 25° ± 1°C. throughout the test. Clean both glass plates thoroughly, removing any ointment from previous tests by rubbing with chloroform and drying. Place the bottom plate, with markings on the under side, in a wooden tray and place on a table in a level position. Break off the tip of the tube, containing the ointment sample, and squeeze 5.0 ± 0.05 gm. of sample in the form of a mound near the center of a piece of "weighing paper" (glassine or equal). Place the "weighing paper" (which must be at least 80 mm. in diameter or length of any side) on the bottom glass so that the mound of sample is approximately in the center of the 5-cm. circle. Place a similar piece of "weighing paper" lightly on the mound of sample. Lower the top plate smoothly on the mound of sample. Place the prepared weight gently on the top plate directly above the 5-cm. circle. Start a stop watch. Beginning 55 seconds after the stop watch was started, make eight light marks, as with India ink, on the top plate, marking the intersection of each of the eight radial lines with the edge of the approximate circle of ointment. The markings shall be made in either clockwise or counter-clockwise order and shall be completed within 65 seconds after the stop watch was started, and within 5 minutes after starting to weigh out the sample. The pressure used in making the markings shall be as light as possible. Remove the weight from the top plate and stop the watch. Measure the distance, in mm., between the two marks located on each of the four diameters. Reserve the ointment on the plate for the grit test. (See par. F–10.) Compute the average of the four distances and report this average, in mm., as the consistency of the ointment.

F–6c. *Rejection.*—If the result of the first determination does not comply with the consistency requirements, additional determinations shall be made, using fresh 5-gm. portions of the sample, until two successive determinations agree to within 2 mm. If the average of these determinations does not comply with the requirements, other tubes from the same portion of the batch may be stored at room temperature (31°C., max.) for an additional time and the determination repeated. If the ointment does not comply with the requirements within a total of 2 months of storage time, the lot of ointment shall be rejected.

F–7. *Heat-stability tests.*—

F–7a. *Method.*—Store the group of five tubes, selected for the heat-stability tests, in an oven at 65° ± 1°C. The oven shall be equipped with an accurate recording thermometer. At the end of 168 ± 3 hours, remove the tubes from the oven and allow to cool to room temperature.

[JAN-O-592]

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

F–7b. *Change in consistency.*—Determine the consistency, using a sample from one of the five tubes which have been heated, within 24 hours after removal from the oven, in accordance with the method prescribed in paragraph F–6b. Reserve the remainder of the ointment in the tube for the test for loss of active chlorine. (See pars. F–7c and F–7d.)

F–7c. *Loss of active chlorine.*—Determine the active chlorine content as described in paragraphs F–5a and F–5b, except that each tube shall be opened, mixed, sampled, and titrated separately. (See par. F–7d.) Calculate the percentage loss of active chlorine as follows:

$$\text{Percentage loss of active chlorine content} = \frac{100(A-B)}{A}$$

where:

A=percentage active chlorine in original (4-tube composite) sample (see par. F–5)

B=percentage active chlorine in heated (1-tube) sample.

F–7d. *Heat stability retests and rejections.*—If any tube of ointment, when tested as prescribed in paragraphs F–7a, F–7b, and F–7c, fails to comply with the requirements for consistency (see par. E–6b) or loss in active chlorine (see par. E–6a), or both, the tube shall be termed a failure. If the single tube tested for change in consistency fails, repeat the test on each of the remaining four tubes. If one of these four tubes fails, the entire lot shall be rejected. However, if no other failure in consistency occurs, determine the loss of active chlorine on each of the five tubes in the group. If two failures (as defined above) occur in consistency or loss of active chlorine, or both, the lot shall be rejected. If one failure is observed in testing the five tubes, an additional five tubes, from the same lot of ointment, shall be retested after heating as prescribed in paragraph F–7a, both for change in consistency (see par. F–7b) and loss of active chlorine (see par. F–7c). If any failure occurs on retest, the lot shall be rejected.

F–8. *Tubing.*—

F–8a. *Cleaning.*—The inspector shall spot check the empty tubes to ascertain compliance with the requirements of paragraph E–7a.

F–8b. *Net weight of filling.*—Weigh and mark three empty tubes. Place a set of three weighed, empty tubes into the filling line at least once every 4 hours. Reweigh the tubes, after filling, and calculate the average net weight of filling. If the average net weight of filling fails to comply with the specified requirements, repeat the test immediately using another set of three weighed tubes. If the average net weight of filling, based on the six tubes tested, does not fall within the prescribed limits (see par. E–7b), the filling machine shall be readjusted and retested, and all tubes filled in the interval between the last satisfactory test and satisfactory readjustment shall be rejected.

F–8c. *Marking.*—Visually examine each filled tube for compliance with the marking requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

F–9. *Leakage.*—Visually examine each filled tube for leakage. All leaking tubes shall be rejected.

F–10. *Grit.*—At the conclusion of the consistency test on the original tube (see par. F–6b), remove the top glass plate. Scrape the ointment from the two "weighing papers" and place on a smooth glass plate. Rub the ointment on the plate with a smooth, steel blade. Consider as grit any particles which tend to give the characteristic tactile sensation experienced when fine sand is ground between steel and glass in the same manner. If grit is found, test another tube in the same way. If the second tube is free from grit, the lot shall be accepted.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *For Army purchases.*—The filled tubes shall be packaged and packed into protective ointment kits and shipments marked as prescribed in Joint Army-Navy Specification JAN–O–591.

G–2. *For Navy purchases.*—

G–2a. *Packaging.*—

G–2a(1). *Unit packaging.*—One tube constitutes one unit. Unless otherwise specified, each tube (1 unit) shall be individually packaged in a folding paperboard carton or set-up paperboard box of appropriate size conforming to the requirements of Joint Army-Navy Specifications JAN–P–120 and JAN–P–133, respectively. Each carton shall be adequately secured to prevent accidental opening.

G–2a(2). *Intermediate packaging.*—Ninety-six packages (96 units) shall be further packaged in a solid fiber box or a corrugated fiber box of appropriate size and design, conforming to the requirements of Federal Specifications LLL–B–636 and LLL–B–631, respectively. Each carton shall be adequately secured to prevent accidental opening.

G–2b. *Packing.*—

G–2b(1). *For domestic shipment.*—Four intermediate packages (384 units) shall be overpacked in a nailed wood box conforming to the requirements of Federal Specification NN–B–621.

G–2b(2). *For overseas shipment.*—Four intermediate packages (384 units) shall be overpacked for type 2 load in a nailed wood box conforming to the requirements of Joint Army-Navy Specification JAN–P–106. Closure and strapping shall conform to the appendix thereto. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, conforming to the requirements of Joint Army-Navy Specification JAN–P–125, for case liners. Seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140.

G–2c. *Marking.*—

G–2c(1). *Interior packages.*—Each interior package shall be marked with the stock number, nomenclature, unit, lot (control) number, quantity, and name of manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

G-2c(2). *Shipping containers.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook, and, in addition, shall bear the lot (control) number.

H. NOTES

H-1. *Uses.*—Ointment covered by this specification is intended for use in the protection of the skin against vesicant-type chemical agents.

H-2. Requests, requisitions, schedules, and contracts or orders should contain the following features:

(a) The title, number, and date of the specification.
(b) The type of packing desired. (See section G.)

H-3. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H-4. If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified in this specification. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures prescribed in this specification.

H-5. Copies of Chemical Corps drawings may be obtained from the Chief of the Chemical Corps, Supply and Procurement Division, Army Chemical Center, Maryland.

H-6. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the Department of the Army will obtain copies of Joint Army-Navy, United States Army and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies.

H-7. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies.

ARMY: CMC.
NAVY: BUSAN.

U. S. GOVERNMENT PRINTING OFFICE: 1949—781467(C-282)

[JAN-C-799]

GPO 889 828

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.

FOLD

EPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
EDGEWOOD ARSENAL
ATTN: SMUEA-QA-WS-S
EDGEWOOD ARSENAL, MARYLAND 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:51Z
Check the source to verify that this is the current version before use.